

Two-View Geometry

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Outline

- Pipeline of 3D Reconstruction
- Triangulation
- Fundamental Matrix
- Essential Matrix

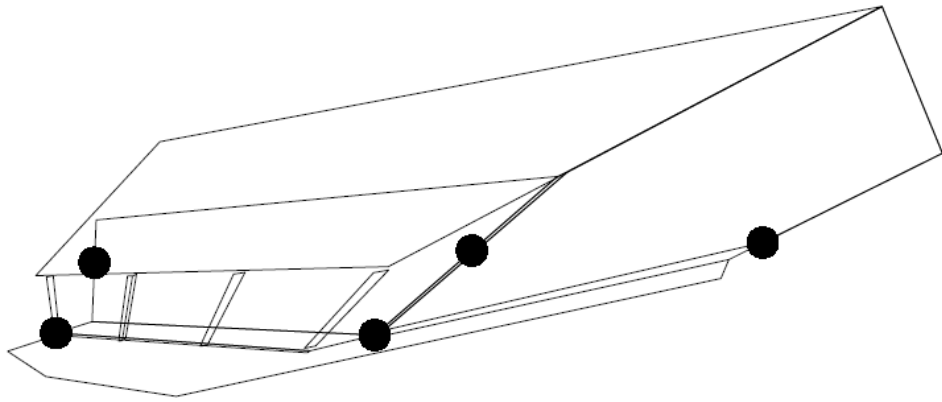
3D Reconstructions

- Compute a projective reconstruction:
 1. Compute the **fundamental matrix** $\mathbf{F} \in \mathbb{R}^{3 \times 3}$ from point correspondences $(\mathbf{x}_i, \mathbf{x}'_i)$ between two images.
 2. Compute the **camera matrices** $\mathbf{P}, \mathbf{P}' \in \mathbb{R}^{3 \times 4}$ from the fundamental matrix.
 3. **Triangulation**: For each correspondence $(\mathbf{x}_i, \mathbf{x}'_i)$, compute $\mathbf{X}_i \in \mathbb{R}^3$ that projects to these two image points.
- Rectify the projective reconstruction to metric
 - Direct method: Use homography.
 - Stratified method: Use affine or metric reconstruction.

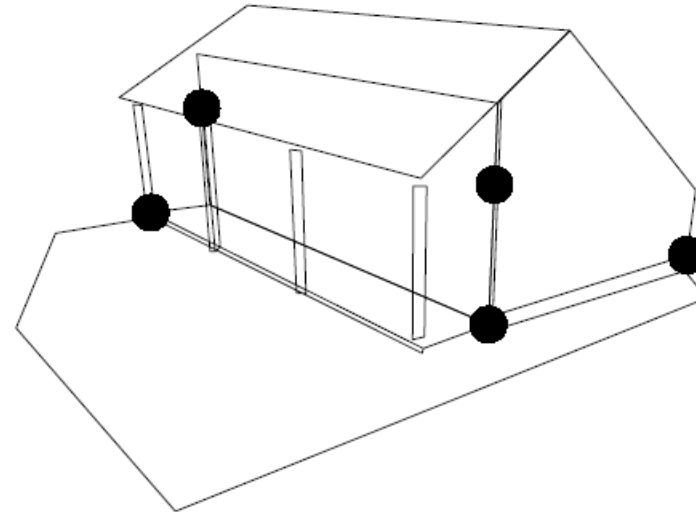
Rectify the Projection Reconstruction



Input Image



Projective Reconstruction



Direct Reconstruction

Triangulation

Given $\mathbf{P}$, $\mathbf{P}'$, (x_i, y_i) , (x'_i, y'_i) for $i = 1, 2, \dots, n$.

Solve (X_i, Y_i, Z_i) for $i = 1, 2, \dots, n$ such that

$$\tilde{\mathbf{m}}_i = \mathbf{P}\tilde{\mathbf{M}}_i \quad \text{and} \quad \tilde{\mathbf{m}}'_i = \mathbf{P}'\tilde{\mathbf{M}}_i$$

The first equation can be written as

$$\begin{bmatrix} \lambda_i x_i \\ \lambda_i y_i \\ \lambda_i \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} X_i \\ Y_i \\ Z_i \\ 1 \end{bmatrix}$$

Solve Triangulation

$$\begin{bmatrix} \lambda_i x_i \\ \lambda_i y_i \\ \lambda_i \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} X_i \\ Y_i \\ Z_i \\ 1 \end{bmatrix} \Rightarrow \begin{cases} \lambda_i x_i = p_{11}X_i + p_{12}Y_i + p_{13}Z_i + p_{14} \\ \lambda_i y_i = p_{21}X_i + p_{22}Y_i + p_{23}Z_i + p_{24} \\ \lambda_i = p_{31}X_i + p_{32}Y_i + p_{33}Z_i + p_{34} \end{cases}$$

$$\Rightarrow \begin{cases} (p_{31}X_i + p_{32}Y_i + p_{33}Z_i + p_{34})x_i = p_{11}X_i + p_{12}Y_i + p_{13}Z_i + p_{14} \\ (p_{31}X_i + p_{32}Y_i + p_{33}Z_i + p_{34})y_i = p_{21}X_i + p_{22}Y_i + p_{23}Z_i + p_{24} \end{cases}$$

$$\Rightarrow \begin{cases} (p_{11} - p_{31}x_i)X_i + (p_{12} - p_{32}x_i)Y_i + (p_{13} - p_{33}x_i)Z_i = p_{34}x_i - p_{14} \\ (p_{21} - p_{31}y_i)X_i + (p_{22} - p_{32}y_i)Y_i + (p_{23} - p_{33}y_i)Z_i = p_{34}y_i - p_{24} \end{cases}$$

Solve Triangulation (cont.)

$$\text{Point 1} \quad \begin{cases} (p_{11} - p_{31}x_i)X_i + (p_{12} - p_{32}x_i)Y_i + (p_{13} - p_{33}x_i)Z_i = p_{34}x_i - p_{14} \\ (p_{21} - p_{31}y_i)X_i + (p_{22} - p_{32}y_i)Y_i + (p_{23} - p_{33}y_i)Z_i = p_{34}y_i - p_{24} \end{cases}$$

$$\text{Point 2} \quad \begin{cases} (p'_{11} - p'_{31}x'_i)X_i + (p'_{12} - p'_{32}x'_i)Y_i + (p'_{13} - p'_{33}x'_i)Z_i = p'_{34}x'_i - p'_{14} \\ (p'_{21} - p'_{31}y'_i)X_i + (p'_{22} - p'_{32}y'_i)Y_i + (p'_{23} - p'_{33}y'_i)Z_i = p'_{34}y'_i - p'_{24} \end{cases}$$

$$\begin{array}{l} \text{Point 1} \\ \text{Point 2} \end{array} \quad \underbrace{\begin{bmatrix} p_{11} - p_{31}x_i & p_{12} - p_{32}x_i & p_{13} - p_{33}x_i \\ p_{21} - p_{31}y_i & p_{22} - p_{32}y_i & p_{23} - p_{33}y_i \\ p'_{11} - p'_{31}x'_i & p'_{12} - p'_{32}x'_i & p'_{13} - p'_{33}x'_i \\ p'_{21} - p'_{31}y'_i & p'_{22} - p'_{32}y'_i & p'_{23} - p'_{33}y'_i \end{bmatrix}}_{\mathbf{A}_i} \underbrace{\begin{bmatrix} X_i \\ Y_i \\ Z_i \end{bmatrix}}_{\mathbf{M}_i} = \underbrace{\begin{bmatrix} p_{34}x_i - p_{14} \\ p_{34}y_i - p_{24} \\ p'_{34}x'_i - p'_{14} \\ p'_{34}y'_i - p'_{24} \end{bmatrix}}_{\mathbf{b}_i}$$

Solve Triangulation (cont.)

Given $\mathbf{P}, \mathbf{P}', (x_i, y_i), (x'_i, y'_i)$ for $i = 1, 2, \dots, n$.

Solve (X_i, Y_i, Z_i) for $i = 1, 2, \dots, n$ such that $\tilde{\mathbf{m}}_i = \mathbf{P}\tilde{\mathbf{M}}_i$ and $\tilde{\mathbf{m}}'_i = \mathbf{P}'\tilde{\mathbf{M}}_i$

- For each correspondence, from the previous slide we have $\mathbf{A}_i\mathbf{M}_i = \mathbf{b}_i$,
where $\mathbf{A}_i \in \mathbb{R}^{4 \times 3}$, $\mathbf{M}_i \in \mathbb{R}^3$, $\mathbf{b}_i \in \mathbb{R}^4$
- Solve $\mathbf{A}_i\mathbf{M}_i = \mathbf{b}_i$ by the method of least squares `np.linalg.lstsq()`.

Outline

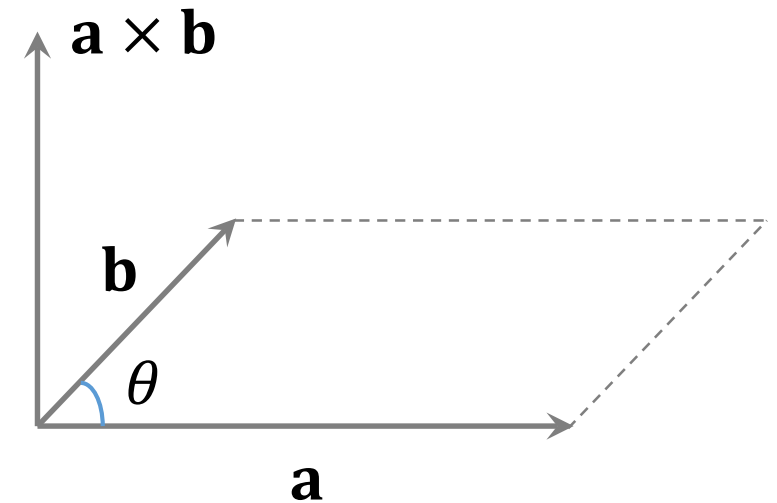
- Pipeline of 3D Reconstruction
- Triangulation
- **Fundamental Matrix**
- Essential Matrix

Cross Product

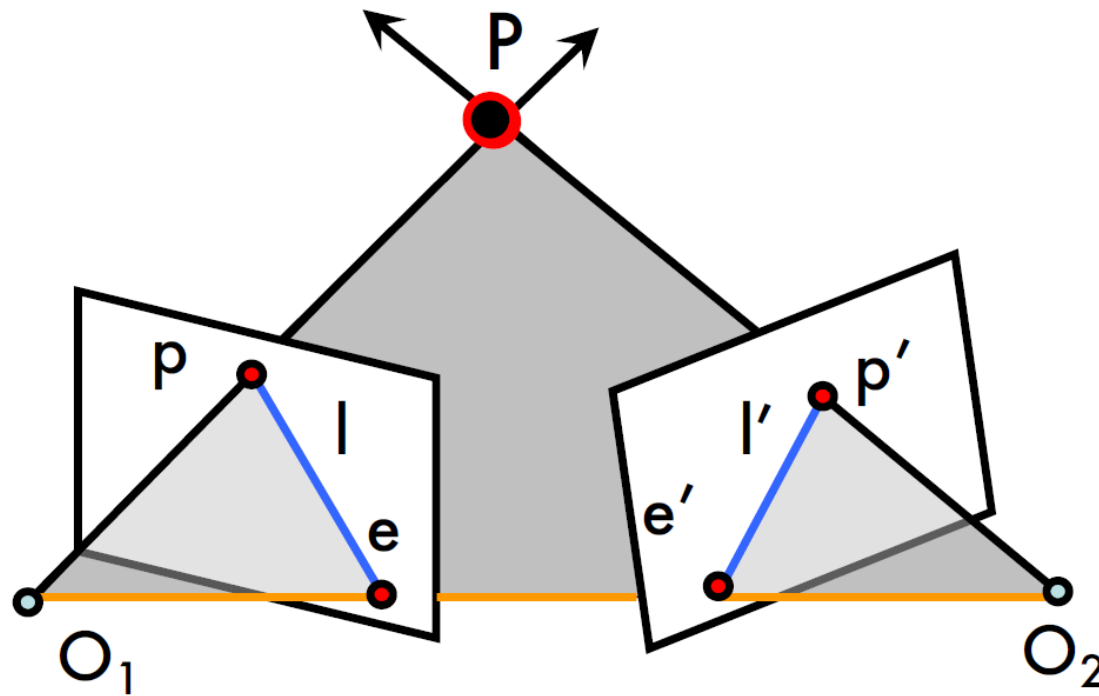
- $\mathbf{a} \times \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \sin(\theta) \mathbf{n}$
- $\mathbf{n}$ is a unit vector perpendicular to the plane containing $\mathbf{a}$ and $\mathbf{b}$, in the direction given by **right-hand rule**.

$$\bullet \mathbf{a} \times \mathbf{b} = \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = [\mathbf{a}_\times] \mathbf{b}$$

- What is the rank of $[\mathbf{a}_\times]$?
- What is $\mathbf{a} \times \mathbf{b}$ if $\mathbf{a}$ and $\mathbf{b}$ are parallel?



Epipolar Constraint: Fundamental Matrix



- Suppose $(\mathbf{p}, \mathbf{p}')$ is a pair of corresponding points in the two images. We have

$$\mathbf{p}'^T \mathbf{F} \mathbf{p} = 0$$

- $\ell' = \mathbf{F} \mathbf{p}$ is the epipolar line associated with $\mathbf{p}$
- $\ell = \mathbf{F}^T \mathbf{p}'$ is the epipolar line associated with $\mathbf{p}'$
- $\mathbf{F} \mathbf{e} = \mathbf{0}$ and $\mathbf{F}^T \mathbf{e}' = \mathbf{0}$
- $\mathbf{F} \in \mathbb{R}^{3 \times 3}$ is of rank two with 7 degrees of freedom

Computing Fundamental Matrix

Given $(x_i, y_i), (x'_i, y'_i)$ for $i = 1, 2, \dots, n$.

Solve $\mathbf{F} \in \mathbb{R}^{3 \times 3}$ such that

$$[x'_i, y'_i, 1] \mathbf{F} [x_i, y_i, 1]^T = 0$$

$$[x'_i, y'_i, 1] \begin{bmatrix} f_1 & f_2 & f_3 \\ f_4 & f_5 & f_6 \\ f_7 & f_8 & f_9 \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} = 0$$

- $\mathbf{F}$ has 7 degrees of freedom.
- At least 8 points are needed to solve linearly for $\mathbf{F}$.
- A non-linear solution is available for 7 point correspondences.

Computing Fundamental Matrix

$$[x'_i, y'_i, 1] \begin{bmatrix} f_1 & f_2 & f_3 \\ f_4 & f_5 & f_6 \\ f_7 & f_8 & f_9 \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} = 0 \text{ for } i = 1, 2, \dots, n$$

$$x_i x'_i f_1 + x'_i y_i f_2 + x'_i f_3 + x_i y'_i f_4 + y_i y'_i f_5 + y'_i f_6 + x_i f_7 + y_i f_8 + f_9 = 0$$

$$\underbrace{\begin{bmatrix} x_1 x'_1 & x'_1 y_1 & x'_1 & x_1 y'_1 & y_1 y'_1 & y'_1 & x_1 & y_1 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_n x'_n & x'_n y_n & x'_n & x_n y'_n & y_n y'_n & y'_n & x_n & y_n & 1 \end{bmatrix}}_{\mathbf{A} \in \mathbb{R}^{n \times 9}} \underbrace{\begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \\ f_5 \\ f_6 \\ f_7 \\ f_8 \\ f_9 \end{bmatrix}}_{\mathbf{f} \in \mathbb{R}^9} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

8-Point Algorithm

- Let $\mathbf{A} \in \mathbb{R}^{n \times 9}$ be the matrix in the previous slide.
- Linear solution:
 - Perform singular value decomposition: $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$
 - Let $\mathbf{f}$ be the last column of $\mathbf{V}$.
 - Reshape vector $\mathbf{f} \in \mathbb{R}^9$ to the fundamental matrix $\mathbf{F} \in \mathbb{R}^{3 \times 3}$.
 - Note that $\mathbf{F}$ does not have rank two.
- Constraint enforcement:
 - Solve $\hat{\mathbf{F}}$ that minimizes $\|\mathbf{F} - \hat{\mathbf{F}}\|$ subject to $\det(\hat{\mathbf{F}}) = 0$.
 - Perform singular value decomposition: $\mathbf{F} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$
 - Let $\mathbf{\Sigma} = \text{diag}(\sigma_1, \sigma_2, \sigma_3)$ with $\sigma_1 \geq \sigma_2 \geq \sigma_3$.
 - Then $\hat{\mathbf{F}} = \mathbf{U} \text{diag}(\sigma_1, \sigma_2, 0) \mathbf{V}^T$.

Normalized 8-Point Algorithm

- Compute $\mathbf{T}$ and $\mathbf{T}'$ for the two images, respectively.
- Normalize image points in the two images.
 - $\mathbf{q}_i = \mathbf{T}\mathbf{p}_i$, $\mathbf{q}'_i = \mathbf{T}'\mathbf{p}'_i$
- Use the 8-point algorithm to compute the fundamental matrix $\hat{\mathbf{F}}$ corresponding to $(\mathbf{q}_i, \mathbf{q}'_i)$.
- De-normalization
 - $\hat{\mathbf{F}}_d = \mathbf{T}'^T \hat{\mathbf{F}} \mathbf{T}$
 - $\hat{\mathbf{F}}_d$ is the fundamental matrix corresponding to the original $(\mathbf{p}_i, \mathbf{p}'_i)$.

Remark

- Suppose $\mathbf{p}_i = (x_i, y_i, 1)$, $\mathbf{p}'_i = (x'_i, y'_i, 1)$ for $i = 1, 2, \dots, n$
- Let $t_x = \frac{1}{n} \sum_{i=1}^n x_i$, $t_y = \frac{1}{n} \sum_{i=1}^n y_i$,
- Let $d = \sum_{i=1}^n \left((x_i - t_x)^2 + (y_i - t_y)^2 \right)^{1/2}$, $s = n\sqrt{2}/d$
- Similarly, we obtain t'_x, t'_y, d', s' by replacing (x_i, y_i) with (x'_i, y'_i) .

$$\mathbf{T} = \begin{bmatrix} s & 0 & -st_x \\ 0 & s & -st_y \\ 0 & 0 & 1 \end{bmatrix}, \mathbf{T}' = \begin{bmatrix} s' & 0 & -s't'_x \\ 0 & s' & -s't'_y \\ 0 & 0 & 1 \end{bmatrix}$$

Normalization

- Transform image coordinates such that matrix $\mathbf{A} \in \mathbb{R}^{n \times 9}$ becomes better conditioned ([pre-conditioning](#)).
- For each image, apply a transformation $\mathbf{T}$ (translation and scaling)
 - $\mathbf{q}_i = \mathbf{T}\mathbf{p}_i$ for $i = 1, 2, \dots, n$ such that
 - The mean of $\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n\}$ is the origin.
 - The mean distance of $\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n\}$ from the origin is $\sqrt{2}$ pixels.
 - The mean distance is defined by $\sum_{i=1}^n \|\mathbf{q}_i\| / n$

Visualize Epipolar Lines

$\mathbf{p}$ is a point in image 1.

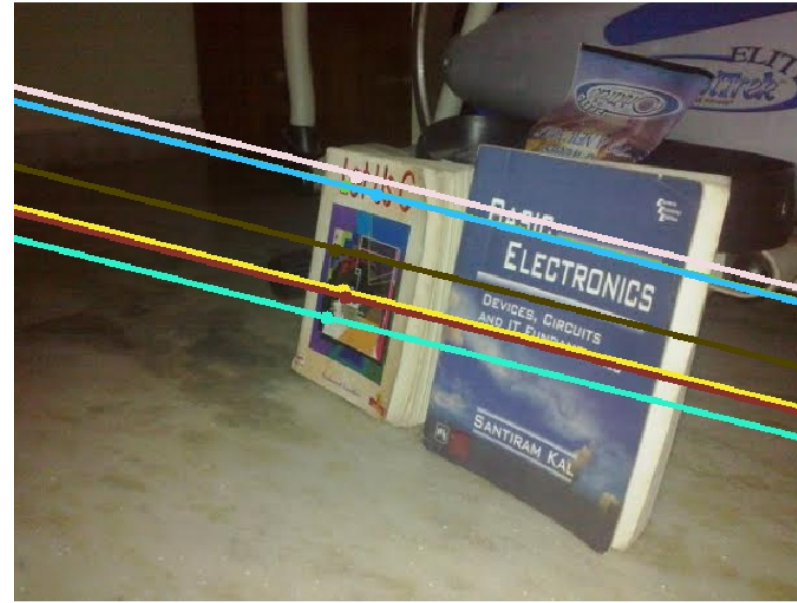


$\mathbf{Fp}$ is the corresponding epipolar line in image 2.



Visualize Epipolar Lines

Where do the epipolar lines intersect at?



Fundamental Matrix is Projective-Invariant

Proposition. If $\mathbf{H} \in \mathbb{R}^{4 \times 4}$ is a projective transformation of 3-space, then the fundamental matrix corresponding to the pairs of camera matrices $(\mathbf{P}, \mathbf{P}')$ and $(\mathbf{PH}, \mathbf{P}'\mathbf{H})$ are the same.

Proof.

Suppose $(\mathbf{x}, \mathbf{x}')$ are matched image points with respect to the pair of camera matrices $(\mathbf{P}, \mathbf{P}')$. Then,

$$\mathbf{x} = \mathbf{PX} \text{ and } \mathbf{x}' = \mathbf{P}'\mathbf{X}$$

where $\mathbf{X}$ represents 3D object points. In view of

$$\mathbf{x} = (\mathbf{PH})(\mathbf{H}^{-1}\mathbf{X}) \text{ and } \mathbf{x}' = (\mathbf{P}'\mathbf{H})(\mathbf{H}^{-1}\mathbf{X})$$

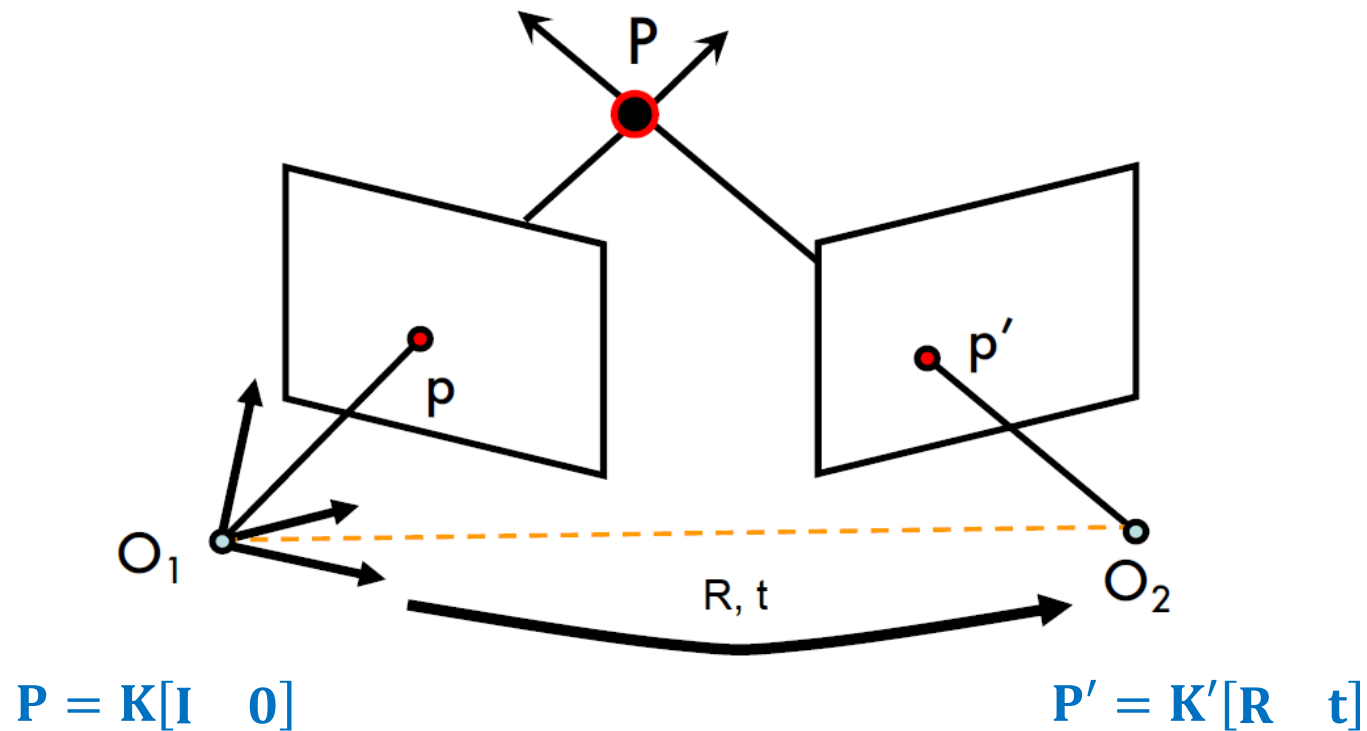
we know $(\mathbf{P}, \mathbf{P}')$ and $(\mathbf{PH}, \mathbf{P}'\mathbf{H})$ give rise to the same fundamental matrix, since the image points remain unchanged.

Remark. However, the 3D object points for $(\mathbf{P}, \mathbf{P}')$ and $(\mathbf{PH}, \mathbf{P}'\mathbf{H})$ are not the same. The former is $\mathbf{X}$, while the latter is $\mathbf{H}^{-1}\mathbf{X}$.

Outline

- Pipeline of 3D Reconstruction
- Triangulation
- Fundamental Matrix
- **Essential Matrix**

Epipolar Constraint: Essential Matrix



- $\mathbf{p}'^T \mathbf{F} \mathbf{p} = 0 \Leftrightarrow (\mathbf{K}'^{-1} \mathbf{p}')^T \mathbf{E} \mathbf{K}^{-1} \mathbf{p}$
- $\mathbf{F} = \mathbf{K}'^{-T} \mathbf{E} \mathbf{K}^{-1}$
- $\mathbf{E} = \mathbf{K}'^T \mathbf{F} \mathbf{K}$
- $\mathbf{E} = [\mathbf{t}_\times] \mathbf{R} = \mathbf{R}[(\mathbf{R}^T \mathbf{t})_\times]$
- $\mathbf{E}$ has only five degrees of freedom
- $\mathbf{E} \in \mathbb{R}^{3 \times 3}$ is an essential matrix if and only if two of its singular values are equal, and the third is zero.

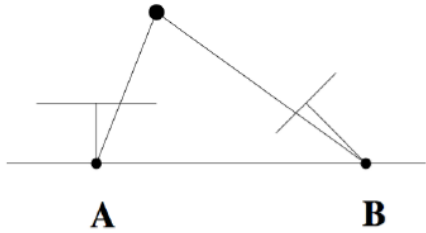
$$\hat{\mathbf{p}}'^T [\mathbf{t}_\times] \mathbf{R} \hat{\mathbf{p}} = 0$$

Longuet-Higgins Equation

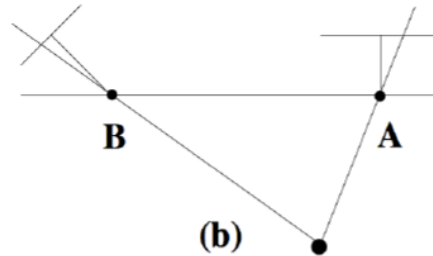
Extracting Camera Matrices from $\mathbf{E}$

- Perform SVD for the essential matrix $\mathbf{E} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$.
- Let $\mathbf{u}_3$ be the last column of $\mathbf{U}$, and $\mathbf{W} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.
- If $\det(\mathbf{U}\mathbf{W}\mathbf{V}^T) < 0$, then replace $\mathbf{W}$ with $-\mathbf{W}$.
- Suppose the first camera matrix is $\mathbf{P} = [\mathbf{I} \quad \mathbf{0}]$.
- There are four possible choices for the second camera matrix $\mathbf{P}' = [\mathbf{R} \quad \mathbf{t}]$:
 - $\mathbf{R}_1 = \mathbf{U}\mathbf{W}\mathbf{V}^T, \mathbf{t}_1 = \mathbf{u}_3$
 - $\mathbf{R}_1 = \mathbf{U}\mathbf{W}\mathbf{V}^T, \mathbf{t}_2 = -\mathbf{u}_3$
 - $\mathbf{R}_2 = \mathbf{U}\mathbf{W}^T\mathbf{V}^T, \mathbf{t}_1 = \mathbf{u}_3$
 - $\mathbf{R}_2 = \mathbf{U}\mathbf{W}^T\mathbf{V}^T, \mathbf{t}_2 = -\mathbf{u}_3$

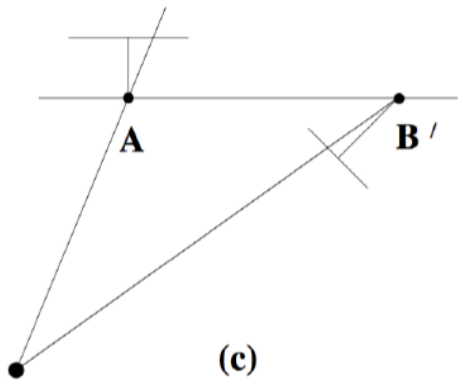
Four Possible Solutions of E



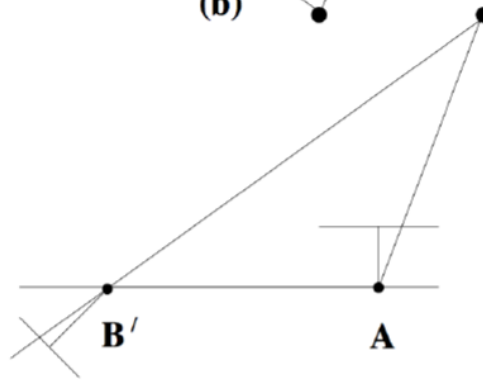
(a)



(b)



(c)



(d)

- Which one do we choose?
- Compute 3D points via triangulation. Valid solutions have positive Z value.
- Valid rotation $\mathbf{R}$ should satisfy $\det(\mathbf{R}) = 1$. Note that $\det(\mathbf{R}) = -1$ means rotation and reflection.